

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSE-AI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No :8(Eight)

Team Members :

Roll No	Student Name
AM.SC.U4AIE25026	PRARTHANA MANJU DEEPAK
AM.SC.U4AIE25003	ANAMIKA MURALI
AM.SC.U4AIE25018	BHUMI AGARWAL
AM.SC.U4AIE25053	GODHARI NAMRATHA

1. Problem Title : Railway Ticket Reservation System

2. Problem Description : Manual reservation systems lead to booking conflicts, long queues, and inefficiency. Automating the process ensures faster booking, secure storage of passenger details, and reduced human error.

3. Identified Entities (Classes)

Class Name	Description
Passenger	Represents individuals booking tickets
TicketManager	Manages train schedules and assists passengers
Admin	Maintains train routes and system data
Train	Represents train details including routes and timings
Reservation	Handles booking, cancellation, and modification of
ReservationSystem	Allocates seats and stores booking details

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(baviours)
Passenger	Name, Age, Contact	SearchTrain(), BookTicket(), CancelTicket()
TicketManager	ManagerID, Name	UpdateSchedule(), VerifyBooking()
Admin	AdminID, Name	AddRoute(), ModifyRoute(), DeleteRoute()
Train	TrainNo, Source,	DisplaySchedule(), CheckAvailability()
Reservation	ReservationID,	ConfirmBooking(), CancelBooking(),
ReservationSystem	Database, SeatMap	AllocateSeat(), StoreDetails(), GeneratePNR()